

Расчетно-графическая работа.

Тема:

Разработка и настройка CI/CD пайплайна для
автоматического развертывания приложений в Kubernetes
кластере с использованием GitLab и Helm

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Задание на оценку 5:

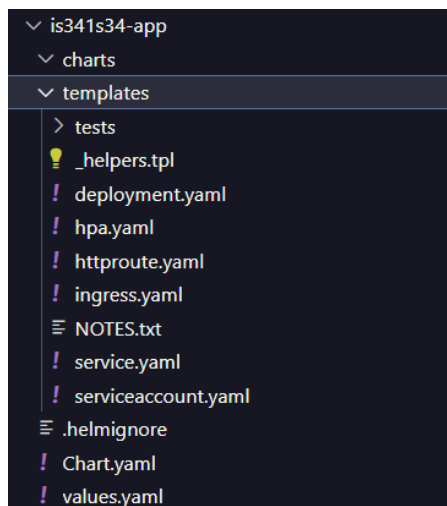
1. Доработать в CI этапы diff и deploy. Для развертывания использовать Helmfile.
2. Создать чарт для развертывания приложения. Привести значения по умолчанию в соответствии с именем проекта.
3. Тег собранного ранее в CI образа передавать через аргументы командной строки.
4. Используя механизм окружений (environment) Helmfile настроить CI таким образом, чтобы разворачивалось два варианта окружения приложений. Все окружения должны работать одновременно. Должна быть возможность запустить деплой каждого из окружений независимо друг от друга. Приложения должны иметь уникальный url и, например, различные значения переменных окружения.
5. Этап deploy должен запускаться по нажатию кнопки.
6. Переменные определенные в environment не должны повторять структуру values. Файлы содержащие environment-переменные не должны использоваться в секциях values.

```
alita@MOBILE:/mnt/c/Users/Кристина/Desktop/classes/reports/ARP/architekura_rp_polyakova_k$ sudo snap install helm --classic
2025-11-01T04:44:05+08:00 INFO Waiting for automatic snapd restart...
helm 3.19.0 from Snapcrafters installed
```

Создали Helm чарт.

```
alita@MOBILE:/mnt/c/Users/Кристина/Desktop/classes/reports/ARP/architekura_rp_polyakova_k$ helm create is341s34-app
```

Настроили структуру проекта:



```
! Chart.yaml X
is341s34-app > ! Chart.yaml
1  apiVersion: v2
2  name: is341s34-app
3  description: A Helm chart for Kubernetes
4  # A chart can be either an 'application' or a 'library' chart.
5  # Application charts are a collection of templates that can be packaged into versioned archives
6  # to be deployed.
7  type: application
8  version: 0.1.0
9  appVersion: "1.16.0"
10
```

Реализация требований:

1. 3 стадии: build → diff → deploy. Параллельные этапы для dev/prod. Deploy по кнопке.

Status	Pipeline	Created by	Stages	Ad
Passed 00:01:32 4 minutes ago	update tag2 #51371 rgr5 0002262c latest branch		✓✓✓	
Blocked 00:00:50	update tag #51369 rgr5 70a388d4 branch		✓✓⊕	
Canceled	update values		✓✓⊕	

deploy-helmfile: [prod]

deploy-helmfile: [dev]

diff-helmfile: [prod]

diff-helmfile: [dev]

build-docker

2. Helmfile для развертывания:

```
helmfile.yaml.gotmpl
1  helmDefaults:
2    wait: true
3    timeout: 600
4    atomic: true
5    waitForJobs: true
6    historyMax: 4
7    createNamespace: false
8
9
10 environments:
11   {{ .Environment.Name }}:
12     missingFileHandler: Warn
13     values:
14       - env/{{ .Environment.Name }}/values.yml
15
16 ---
17
18 releases:
19   - name: is341s34-app-{{ .Environment.Name }}
20     chart: ./is341s34-app
21     version: 0.1.0
22     namespace: "is341"
23     values:
24       - values/is341s34-app.yaml.gotmpl
```

3. Тег образа через аргументы:

```
helmfile -e $ENVIRONMENT \
  --namespace is341 \
  --state-values-set imageRepository="$HAWK_REGISTRY_IMAGE" \
  --state-values-set imageTag="$IMAGE_TAG" \
  --state-values-set replicaCount=1 \
  apply
```

4. Уникальные URL и разные переменные:

Dev: is341s34-app-dev.hawk-dev.csc.sibsutis.ru +
 NODE_ENV=development

Prod: is341s34-app.hawk-dev.csc.sibsutis.ru +
 NODE_ENV=production

5. Переменные определенные в environment не должны повторять структуру values.

The screenshot displays a code editor with two files open. The top file is `values.yaml` for the `dev` environment, showing the following content:

```
1 appDomain: "is341s34-app-dev.hawk-dev.csc.sibsutis.ru"
2 nodeEnvironment: "development"
```

The bottom file is `is341s34-app.yaml.gotmpl`, which is a Helm chart template. It includes conditional rendering for the `ingress` and `env` sections, and a `commonLabels` section. The template uses the variables defined in the `values.yaml` file:

```
1 {{- if .Values.appDomain }}
2 ingress:
3   enabled: true
4   className: "traefik-dev"
5   hosts:
6     - host: {{ .Values.appDomain | quote }}
7     paths:
8       - path: /
9       pathType: Prefix
10 {{- end }}
11
12 {{- if .Values.nodeEnvironment }}
13 env:
14   NODE_ENV: {{ .Values.nodeEnvironment | quote }}
15 {{- end }}
16
17 commonLabels:
18   owner: is341s34
19
20 {{- if .Values.replicaCount }}
21 replicaCount: {{ .Values.replicaCount }}
22 {{- end }}
23
24 image:
25   repository: "cr.hawk.csc.sibsutis.ru/hawk-dev/is341s34-architekura-rp-polyakova-k"
26   tag: {{ .Values.imageTag | default "latest" | quote }}
27   pullPolicy: IfNotPresent
28
```

Дополнительное приложение:

```
is341s34-app > ! values.yaml
1 replicaCount: 1
2
3 image:
4   repository: ""
5   pullPolicy: IfNotPresent
6   tag: ""
7
8 nameOverride: ""
9 fullnameOverride: ""
10
11 serviceAccount:
12   create: false
13
14 podSecurityContext:
15   runAsNonRoot: true
16   runAsUser: 1000
17
18 securityContext:
19   allowPrivilegeEscalation: false
20   capabilities:
21     drop:
22       - ALL
23
24 service:
25   type: ClusterIP
26   port: 80
27   targetPort: http
28
29 ingress:
30   enabled: true
31   className: "traefik-dev"
32   annotations: {}
33   hosts: []
34   tls: []
35
36 env: {}
37
38 resources:
39   requests:
40     cpu: 50m
41     memory: 64Mi
42   limits:
43     cpu: 100m
44     memory: 128Mi
45
46 autoscaling:
47   enabled: false
48
49 nodeSelector: {}
50
51 tolerations: []
52
53 affinity: {}
```

```
is341s34-app > templates > ! deployment.yaml
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: {{ include "is341s34-app.fullname" . }}
5   namespace: is341
6   labels:
7     {{ include "is341s34-app.labels" . | nindent 4 }}
8 spec:
9   replicas: {{ .Values.replicaCount }}
10  selector:
11    matchLabels:
12      {{ include "is341s34-app.selectorLabels" . | nindent 6 }}
13  template:
14    metadata:
15      labels:
16        {{ include "is341s34-app.selectorLabels" . | nindent 8 }}
17        owner: is341s34
18    spec:
19      containers:
20        - name: {{ .Chart.Name }}
21          image: "{{ .Values.image.repository }}:{{ .Values.image.tag }}"
22          imagePullPolicy: {{ .Values.image.pullPolicy }}
23          ports:
24            - name: http
25              containerPort: 3000
26              protocol: TCP
27          {{- with .Values.env }}
28            env:
29              {{- range $key, $value := . }}
30                - name: {{ $key }}
31                  value: {{ $value | quote }}
32              {{- end }}
33          {{- end }}
34          livenessProbe:
35            httpGet:
36              path: /random-number
37              port: 3000
38            initialDelaySeconds: 30
39            periodSeconds: 10
40            timeoutSeconds: 5
41          readinessProbe:
42            httpGet:
43              path: /random-number
44              port: 3000
45            initialDelaySeconds: 5
46            periodSeconds: 5
47            timeoutSeconds: 3
48          resources:
49            {{- toYaml .Values.resources | nindent 12 }}
50          securityContext:
51            allowPrivilegeEscalation: false
52            runAsNonRoot: true
53            runAsUser: 1000
54          capabilities:
55            drop:
56              - ALL
```

```

is341s34-app > templates > ! ingress.yaml
1  {{- if .Values.ingress.enabled -}}
2  apiVersion: networking.k8s.io/v1
3  kind: Ingress
4  metadata:
5    name: {{ include "is341s34-app.fullname" . }}
6    namespace: is341
7    labels:
8      {{- include "is341s34-app.labels" . | nindent 4 }}
9    owner: is341s34
10   {{- with .Values.ingress.annotations }}
11   annotations:
12     {{- toYaml . | nindent 4 }}
13   {{- end }}
14   spec:
15     {{- with .Values.ingress.className }}
16     ingressClassName: {{ . }}
17     {{- end }}
18     {{- if .Values.ingress.tls }}
19     tls:
20       {{- range .Values.ingress.tls }}
21       - hosts:
22         {{- range .hosts }}
23         - {{ . | quote }}
24         {{- end }}
25         secretName: {{ .secretName }}
26       {{- end }}
27     {{- end }}
28     rules:
29       {{- range .Values.ingress.hosts }}
30       - host: {{ .host | quote }}

```

```

is341s34-app > templates > ! ingress.yaml
14  spec:
19  tls:
21    - hosts:
22      {{- range .hosts }}
23      - {{ . | quote }}
24      {{- end }}
25      secretName: {{ .secretName }}
26    {{- end }}
27  {{- end }}
28  rules:
29    {{- range .Values.ingress.hosts }}
30    - host: {{ .host | quote }}
31      http:
32        paths:
33          {{- range .paths }}
34          - path: {{ .path }}
35            {{- with .pathType }}
36            pathType: {{ . }}
37            {{- end }}
38          backend:
39            service:
40              name: {{ include "is341s34-app.fullname" $ }}
41              port:
42                number: {{ $.Values.service.port }}
43            {{- end }}
44          {{- end }}
45    {{- end }}

```

```

is341s34-app > templates > ! service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: {{ include "is341s34-app.fullname" . }}
5    namespace: is341
6    labels:
7      {{- include "is341s34-app.labels" . | nindent 4 }}
8    owner: is341s34
9  spec:
10   type: {{ .Values.service.type }}
11   ports:
12     - port: {{ .Values.service.port }}
13       targetPort: 3000
14       protocol: TCP
15       name: http
16   selector:
17     {{- include "is341s34-app.selectorLabels" . | nindent 4 }}

```

.gitlab-ci.yml:

```
stages:
    build
    diff-helmfile
    deploy-helmfile

build-docker:
stage: build
image: docker:cli
services:
    docker:dind
before_script:
    docker login -u "$CI_REGISTRY_USER" -p
"$CI_REGISTRY_PASSWORD" "$CI_REGISTRY"
    docker login -u "$HAWK_REGISTRY_USER" -p
"$HAWK_REGISTRY_PASSWORD" "$HAWK_REGISTRY"
    docker context create ci
    docker buildx create --name multiarch --use --driver
    docker-container ci
script:
```

```

|
if [ -n "$CI_COMMIT_TAG" ]; then
    IMAGE_TAG=$CI_COMMIT_TAG
else
    IMAGE_TAG=$CI_COMMIT_SHORT_SHA
fi
docker buildx build --provenance=false --push --platform
linux/arm64,linux/amd64 --tag
$CI_REGISTRY_IMAGE:$IMAGE_TAG --tag
$HAWK_REGISTRY_IMAGE:$IMAGE_TAG .
rules:
  if: $CI_COMMIT_TAG
  when: always
  if: $CI_COMMIT_BRANCH
  when: manual

.helmfile-pattern:
image: registry.csc.sibsutis.ru/csc-public/internal-container-
images/k8s-tools:latest
before_script:
  echo "$KUBECONFIG_CONTENT" > /tmp/kubeconfig
  export KUBECONFIG=/tmp/kubeconfig
  export HELMFILE_FILE_PATH=helmfile.yaml.gotmpl
rules:
  if: $CI_COMMIT_TAG
  when: manual
  if: $CI_COMMIT_BRANCH
  when: manual
dependencies:
  build-docker

diff-helmfile:
extends: .helmfile-pattern
stage: diff-helmfile
parallel:
matrix:
  ENVIRONMENT: [dev, prod]
script:
  |
  if [ -n "$CI_COMMIT_TAG" ]; then
      IMAGE_TAG=$CI_COMMIT_TAG
  else
      IMAGE_TAG=$CI_COMMIT_SHORT_SHA
  fi
  echo "Дифф для окружения: $ENVIRONMENT"
  echo "образ: $HAWK_REGISTRY_IMAGE:$IMAGE_TAG"

  helmfile -e $ENVIRONMENT \
    --namespace is341 \

```



```

        --state-values-set
        imageRepository="$HAWK_REGISTRY_IMAGE" \
        --state-values-set imageTag="$IMAGE_TAG" \
        --state-values-set replicaCount=1 \
        diff

deploy-helmfile:
extends: .helmfile-pattern
stage: deploy-helmfile
parallel:
matrix:
    ENVIRONMENT: [dev, prod]
script:
    |
    if [ -n "$CI_COMMIT_TAG" ]; then
        IMAGE_TAG=$CI_COMMIT_TAG
    else
        IMAGE_TAG=$CI_COMMIT_SHORT_SHA
    fi

    echo "Деплой в окружение: $ENVIRONMENT"
    echo "Используется образ:
    $HAWK_REGISTRY_IMAGE:$IMAGE_TAG"

    helmfile -e $ENVIRONMENT \
        --namespace is341 \
        --state-values-set
        imageRepository="$HAWK_REGISTRY_IMAGE" \
        --state-values-set imageTag="$IMAGE_TAG" \
        --state-values-set replicaCount=1 \
        apply

    if [ "$ENVIRONMENT" = "dev" ]; then
        echo "Dev приложение доступно по: http://is341s34-
        app-dev.hawk-dev.csc.sibsutis.ru"
    else
        echo "Prod приложение доступно по: http://is341s34-
        app.hawk-dev.csc.sibsutis.ru"
    fi

```